Gradient function graphs

1

a 1

b 1

c A linear function has a constant gradient.

2

a The gradient function is positive when $y = x^2$ is increasing, and negative when $y = x^2$ is decreasing.

b Positive

c The gradient of the tangent line increases at a constant rate.

d Negative

e The gradient of the tangent line increases at a constant rate.

f Linear

3

a Positive

b The gradient of the tangent line increases at an increasing rate.

c Positive

d The gradient of the tangent line decreases at an decreasing rate.

e Quadratic

4

a Positive

b The gradient of the tangent line increases at an increasing rate.

c Negative

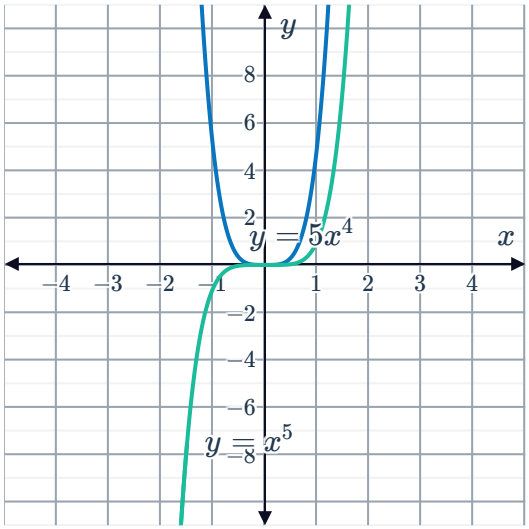
d The gradient of the tangent line increases at a decreasing rate.

e Cubic

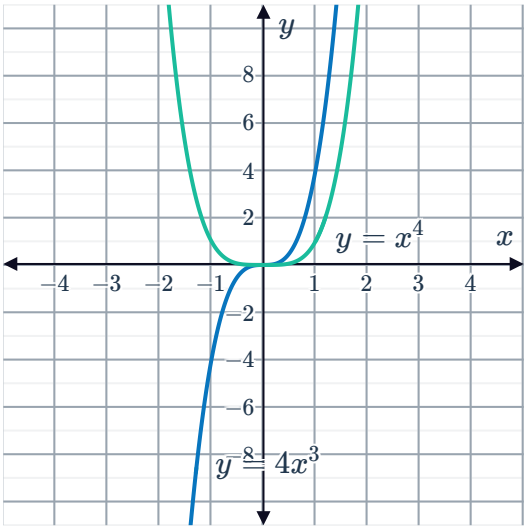
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5

a



b



c Odd

d Even

Algebraic differentiation

6

a

i Degree 1 **ii** Degree 2 **iii** Degree 3

b $n-1$

7

a $\frac{dy}{dx} = 7x^6$ **b** $\frac{dy}{dx} = 9x^8$ **c** $\frac{dy}{dx} = 5x^4$ **d** $\frac{dy}{dx} = 0$

Applications

8 $f'(2) = 32$

9

a 9

b $y = 9x - 8$

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10

a

Graph	y-intercept of the tangent line	Gradient of the tangent
$y = x^2$	$(0, -1)$	2
$y = x^3$	$(0, -2)$	3
$y = x^4$	$(0, -3)$	4
$y = x^5$	$(0, -4)$	5

b n

c No, because then the gradient of the tangent at $x = 1$ would be 1.

11

a False **b** True **c** False **d** True

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